



△ **Leaf-shaped** carnelian cabochon

Chalcedony

Chalcedony is a compact variety of quartz, composed of crystals that are microscopic (microcrystalline) or too small even for a standard optical microscope (cryptocrystalline). It forms in cavities, cracks, and by replacement when low-temperature, silica-rich waters percolate through preexisting rocks, in particular volcanic rocks. It is relatively porous, and much chalcedony on the commercial market has been dyed to enhance or colour it artificially. Chalcedonies of all kinds have been used as gems, beads, carvings, and in seals for thousands of years. The earliest stone tools were generally made of some form of chalcedony.

Varieties

Pure chalcedony is white. However, when trace elements or microscopic inclusions of other minerals occur, it can yield a range of colours. Many of these have their own variety names: chalcedony that shows distinct banding is called agate; blood-red to reddish-orange translucent chalcedony coloured by inclusions of iron oxide is known as carnelian; bloodstone is dark, opaque green coloured by traces of iron silicates and with patches of bright red jasper. Chrysoprase is a translucent apple-green variety, coloured by nickel; sard is light to dark brown chalcedony, while sardonyx is colour-banded sard; jasper, chert, and flint are opaque, fine-grained or dense, impure varieties of cryptocrystalline quartz.

Specification

Chemical name Silicon dioxide | **Formula** SiO_2 | **Colours**

All colours | **Structure** Hexagonal/trigonal | **Hardness** 7

SG 2.65 | **RI** 1.54–1.55 | **Lustre** Vitreous



Cabochon



Cameo



Slab



Locations

1 USA **2** Peru **3** Morocco **4** Scotland **5** Netherlands
6 Czech Republic **7** Poland **8** Madagascar **9** Sri Lanka
10 Myanmar **11** Russia

Key pieces



Ancient Egyptian gold pectoral | The lavish gold funerary pectoral of Pharaoh Psusennes I (3rd Pharaoh, 21st Dynasty), from around 1040–996 BCE, is decorated with inlays of red jasper and lapis lazuli.



Aztec knife | The blade of this highly decorated Aztec sacrificial knife is chipped from fine flint, with a mosaic handle of turquoise, coral, and jet. It is thought to originate from around the 15th to 16th centuries.



Chalcedony cup | Combining superb lapidary and enamelling work, this antique cup has been delicately carved from waxy grey chalcedony material and set with gold trimming. The enamel decoration is particularly fine on the handles.